05. THE CELL: METABOLIC FIELDS AND DIFFERENTIATION

DURATION : 11:37 STUDY SHEET

COURSE SUMMARY

This session on biodynamic embryology teaches the fundamentals of cellular development, explaining the trajectory from stem cell to differentiation. With essential concepts such as cleavage, polarizing substances, and metabolic fields, students will understand how cells communicate and interact. Emphasis is placed on the importance of polarization in oocyte formation, as well as on the types of cellular communication, illustrating concrete examples to enrich the understanding of embryogenesis.

BIODYNAMIC EMBRYOLOGY: UNDERSTANDING CELLULAR DEVELOPMENT

This biodynamic osteopathy course explores the fascinating mechanisms of embryology, based on the concepts of **Marc Damoiseaux**. We will delve into the heart of the cell, from its birth to its differentiation, including the forces that shape it.

I. FROM STEM CELL TO POLARISATION: LIFE'S FIRST STEPS

Everything begins with a **stem cell**, the **germarium**, in women. This cell is the starting point for the formation of **oocytes**.

A. CLEAVAGE AND THE FORMATION OF SPECIALISED CELLS

The initial process is **cleavage**, a particular cell division where the cell folds.

Imagine a cell dividing to form two, but unevenly. One of them, stronger, is spared.

From this small cell, a group of cells will emerge. Observing this group after cleavage, we distinguish:

- A large cell, the **oocyte**.
- Smaller cells around it, called **follicular cells**.
- Larger cells, the **nurse cells**.

Thus, the germarium, a stem cell, gives rise to a multitude of cells: **follicular cells** (small, surrounding) and **nurse cells** (large, accompanying).

B. THE ORIGIN OF CELL LINEAGES

It is crucial to understand the origin of these cells:

- Some come from the **germline** (from the mother).
- Others, like **nurse cells**, arise from developmental force.

C. THE ROLE OF MOLECULES AND POLARISATION

Nurse cells and **follicular cells** release small molecules.

These molecules will arrange themselves in the oocyte's **cytoskeleton**, creating specific concentration axes.

In other words, if elements are concentrated on one side, the metabolic concentration will be higher in that area. This is known as **"Wolpert's flag"**.

This concentration implies a "plus" pole and a "minus" pole.

A similar phenomenon occurs along another axis, the **apico-basal axis**. The oocyte polarises, creating axes of force concentration to capture electromagnetic energy. It is therefore oriented from within.

II. CELLULAR CENTRES AND DIVISION

When observing a cell, two main centres are identified:

- A **pure cellular centre**, with the nucleus.
- A **structural centre**, composed of centrioles and microtubules.

A. CELL MULTIPLICATION

The cell will cleave and multiply. This process involves the participation of the structural centre and the cellular centre.

The cellular centre is responsible for releasing **DNA** in the form of photocopies. The "books from the library" (the original DNA) are never taken out; copies are made.

Do you remember the **equatorial plate** during cell division? That's where it all begins. In our work, we will seek to follow this polarity at each stage, even if it takes different forms.

III. THE METABOLIC FIELD: A KEY CONCEPT

To understand embryo development, it is essential to grasp the concept of the **metabolic field**.

A. DEFINITION OF THE METABOLIC FIELD

A metabolic field is a **space** in which the cell evolves. It is a current, dynamic space.

B. Different types of cell communication

To illustrate the interactions within this field, Marc Damoiseaux uses striking analogies:

- **Autocrine communication:** Imagine you have an urgent need and the wind blows everything back onto you. This is communication of the cell with itself.
- **Paracrine communication:** If several of you are "peeing" together and the wind disperses the odours onto your neighbours. This is communication between nearby cells.
- **Endocrine communication:** If you "pee in the gutter" (the circulatory system), the message is transported over a distance.
- **Neurocrine communication:** Similar to paracrine, but over a distance. Neuronal cells create large axons to communicate far away.
- **Telocrine communication:** Means "at a distance".

C. The metabolic field as a structured environment

The metabolic field is an environment around the cell that will become structured.

Take the example of **sauerkraut**: if someone enters a room where you have just "degassed", they see nothing, but they immediately perceive your metabolic field. There is a space around you containing small substances.

This field will structure, organise, and complexify itself.

IV. Differentiation, multiplication, and limits

Differentiation is a fundamental process of development.

A. Cell differentiation

Differentiation involves:

- A **change in position**.
- A **change in shape**.
- A **change in structure and function**.

Two different functions imply two different forms. The word **"biodynamic"** means the dynamics of life in its development.

B. Boundaries and cellular response

Blechmit, a theorist, speaks of different metabolic fields. The metabolic field is not just an environment; it is structured.

Imagine a cell in its space. As long as it stays in this space, everything is fine. But if it touches a boundary, this can cause:

- **Multiplication**.

- Then multiplication with **differentiation**.

This depends on the impact in spatio-temporal space and the genetic response to the environment.

C. TRAUMA AND APOPTOSIS

There are limits beyond which the system can multiply or differentiate.

Traumas, or shocks, can cause profound differentiations in the system. It is crucial to recognise these "figures" (these signals) to help the body restructure itself.

The body possesses a power of **self-regulation**. The environment and the forces acting on the cells can cause division and multiplication.

For differentiation to occur, the pressure must be greater in a given environment. There is a clear distinction between simple multiplication and multiplication with differentiation.

If a cell undergoes too much stress, it dies. This process is called **cellular apoptosis**.

The body uses multiplication, differentiation, and apoptosis to develop and grow. If a group of cells receives too much compressive mechanical stress, only a **cellular exudate** will remain.

This course has allowed us to lay the foundations for understanding biodynamic embryology, highlighting the importance of **metabolic fields**, **cellular communications**, and the forces that guide the development of life.

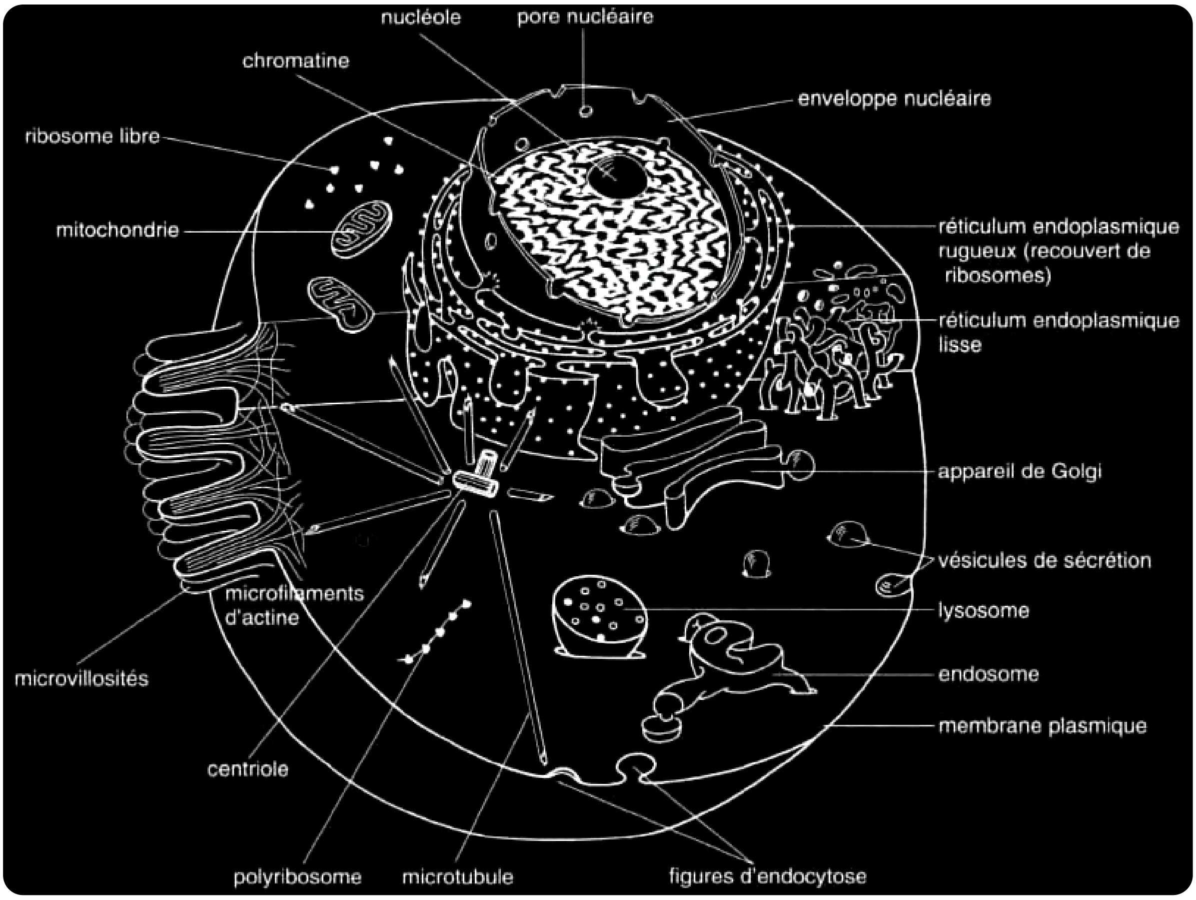

FIG. — *Eukaryotic cell*

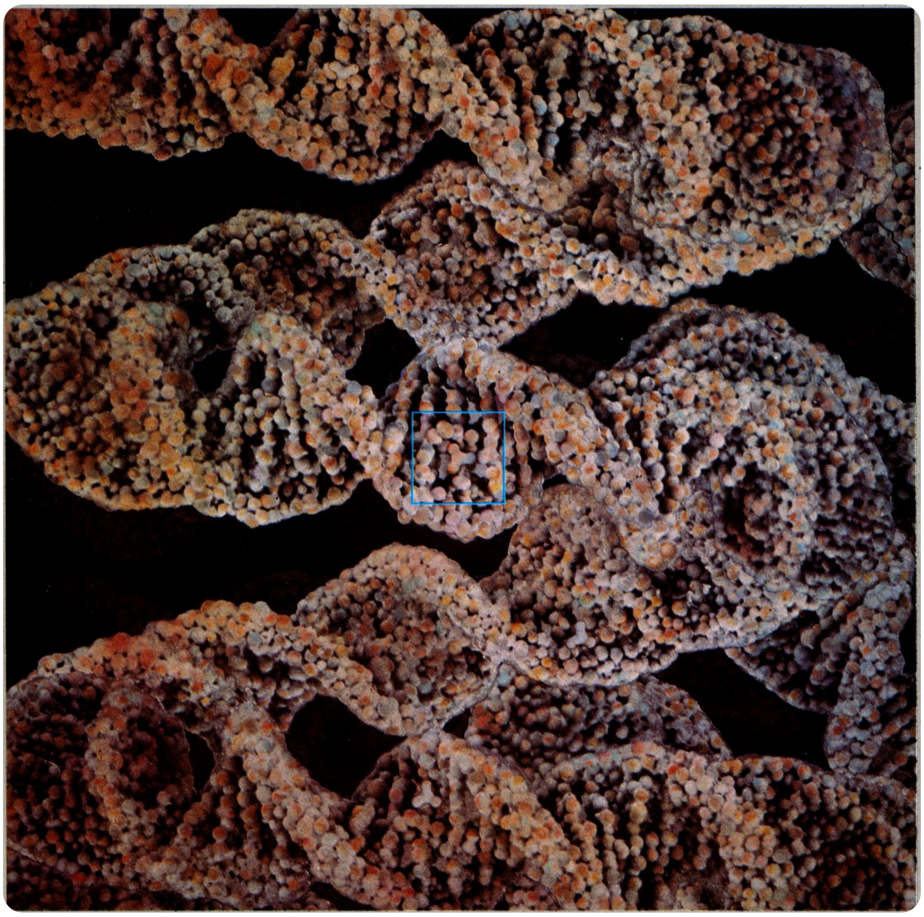

FIG. — Granular histological structure